Bond Duration | Five practice questions

LECTURE 12 · FINA 402 · Luis Ceballos

Try all five questions before reading the solutions. These use different numbers from the lecture. Use first-order duration for yield shocks and keep full precision until the final answer.

1. Macaulay from present values

Cash flows arrive at 0.5, 1.0, 1.5 and 2.0 years. Their present values are $25, $25, $25 and $925. What is Macaulay duration?

A. 1.925 years **B.** 2.000 years **C.** 1.250 years **D.** 0.9625 years

2. Portfolio Macaulay duration

Market values are $300,000 in Bond A (duration 2 years), $200,000 in B (4 years), and $500,000 in C (7 years). What is portfolio Macaulay duration?

A. 4.333 years **B.** 4.900 years **C.** 5.500 years **D.** 13.000 years

3. Modified duration and a price change

A bond position is worth $500,000. Macaulay duration is 4.12 years; annual YTM is 6%, compounded semiannually. Yields increase by 50 bp. Which modified duration and approximate dollar change are correct?

A. 4.12; -$10,300 **B.** 4.00; +$10,000 **C.** 4.00; -$10,000 **D.** 3.887; -$9,717

4. Bank equity modified duration

A hypothetical bank has assets $150bn and liabilities $130bn. Their Macaulay durations are 3.12 and 1.04 years. Both use an 8% annual yield with semiannual compounding. What is equity modified duration?

A. 2.00 **B.** 3.00 **C.** 16.64 **D.** 16.00

5. Equity after a rate shock

For the bank in Question 4, yields rise by 75 bp. What are the approximate equity loss and new equity value?

A. Loss $3.375bn; new equity $16.625bn **B.** Loss $2.400bn; new equity $17.600bn **C.** Loss $4.350bn; new equity $15.650bn **D.** Loss $0.975bn; new equity $19.025bn

Worked solutions

1. Macaulay from present values - Answer A

Step 1. The present values are already given, so do not discount them again. Sum them to find price: $25 + $25 + $25 + $925 = $1,000. The weights are 0.025, 0.025, 0.025 and 0.925.

Step 2. Use time in years

Multiply each payment time by its present-value weight. Most value arrives at maturity, so the result is close to two years.

$$D_{Mac} = 0.5(0.025) + 1(0.025) + 1.5(0.025) + 2(0.925) = 1.925 \; years$$

2. Portfolio Macaulay duration - Answer B

Step 1. Total market value is $1,000,000. The weights are 30%, 20% and 50%; they are not equal just because there are three bonds.

$$D_{Mac,P} = 0.30 \times 2 + 0.20 \times 4 + 0.50 \times 7 = 4.90 \; years$$

The result describes weighted cash-flow timing. It is not a guarantee that the portfolio recovers its purchase price after 4.9 years.

3. Modified duration and a price change - Answer C

Step 1. The annual YTM is 6%, so the semiannual rate is 3%. Convert the given Macaulay duration.

$$D_{mod} = \frac{4.12}{1 + 0.06/2} = 4.00$$

Step 2. Convert the shock and estimate the dollar change

50 bp is 0.005. A positive yield change produces a negative price change for a positive-duration bond.

$$\Delta P \approx -4 \times 500{,}000 \times 0.005 = -\$10{,}000$$

The percentage loss is 2%, and the approximate new market value is $490,000.

Bank equity: build the sensitivity

4. Bank equity modified duration - Answer D

Step 1. Identify the equity cushion before calculating the sensitivity.

$$E = A - L = 150 - 130 = 20 \text{ billion dollars}$$

Step 2. Convert both given durations

They are Macaulay durations, so divide each by 1 + 0.08/2 = 1.04. Do not plug them directly into the price-change formula.

$$D_{mod,A} = \frac{3.12}{1.04} = 3, \qquad D_{mod,L} = \frac{1.04}{1.04} = 1$$

Step 3. Find net dollar duration

Assets expose equity to rate losses. The fall in liability value offsets some of that exposure, so subtract its dollar duration.

$$A \times D_{mod,A} - L \times D_{mod,L} = 150 \times 3 - 130 \times 1 = 320$$

Step 4. Divide by equity, not assets

The residual exposure belongs to the $20bn equity position. That is the correct denominator.

$$D_{mod,E} = \frac{150 \times 3 - 130 \times 1}{20} = 16$$

Economic interpretation: for a small +1 percentage-point parallel yield shock, equity would fall approximately 16%. That is much larger than the 3% asset decline. The bank is leveraged, and the same net dollar loss is concentrated in a small equity base.

Bank equity: apply the rate shock

5. Equity after a rate shock - Answer B

Step 1. Write 75 bp as 0.0075. Use the modified durations from Question 4: 3 for assets and 1 for liabilities.

Step 2. Revalue each side approximately

All dollar amounts remain in billions.

$$\Delta A \approx -3 \times 150 \times 0.0075 = -3.375$$

$$\Delta L \approx -1 \times 130 \times 0.0075 = -0.975$$

Step 3. Take the difference

A smaller liability value benefits the residual equity claim. Subtract the negative liability change.

$$\Delta E = \Delta A - \Delta L = -3.375 - (-0.975) = -2.400$$

Step 4. Check with equity modified duration

This gives the same result and confirms the signs and denominator.

$$\Delta E \approx -16 \times 20 \times 0.0075 = -2.400$$

Step 5. Calculate the percentage loss and new equity

The $2.4bn loss is 12% of initial equity. Use initial equity of $20bn in the denominator.

$$\frac{\Delta E}{E} = \frac{-2.4}{20} = -12\%, \qquad E_{new} = 20 - 2.4 = 17.6$$

Balance-sheet check: new assets = $146.625bn; new liabilities = $129.025bn; their difference is $17.600bn. Duration is a local approximation, not an exact forecast for a bank with changing cash flows, spreads or deposit behavior.